Bhavya Kumar

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Technical Skills

Languages: Python, C, C++, C#, JavaScript, Racket, SQL
AI/ML: LLM Integration (Groq, DeepSeek-R1), LangChain, Prompt Engineering, NLP Intent Matching
Frameworks & Libraries: React, Node.js, Express, Supabase, dnd-kit, REST APIs
Tools: Git, GitHub, Vercel, VS Code, Claude Code, Arduino IDE

Education

University of Waterloo — Waterloo, ON
Honours Bachelor of Mathematics (Co-op) — *Sept. 2025 – April. 2030 (expected)*

- **Major:** Applied Mathematics with Scientific Computing and Scientific Machine Learning.
- **Award:** President's Scholarship.
- **Relevant Coursework:** Data Types and Structures, Probability, Algorithm Design in C, Linear Algebra II, Calculus III, Functional Programming in Racket.

Projects

Software Factory - SuperPlane Hackathon (NYC) - *SuperPlane, LLM Agents* — In-person, June 28, 2026

- Built an autonomous "software factory" on SuperPlane's visual Canvas that turns a rough natural-language feature request into a working proof of concept with minimal human input, orchestrating LLM agents across speccing, coding, and verification stages.
- Designed a multi-stage agent pipeline with validation gates, where each stage confirms the previous step actually succeeded before proceeding, containing cascading failures from unreliable LLM output.
- Automated end-to-end delivery so each generated PoC is committed and deployed to a live preview environment on Render, with a testable link on the pull request, validating the pipeline against real open issues from the SuperPlane repository.

Wyag - Git Version Control System - *Python* | GitHub — May. 2026

- Engineered a fully functional Git client from scratch, implementing a Merkle DAG data structure and a content-addressed object store using SHA-1 hashing and zlib compression.
- Overcame binary parsing complexities by developing a custom parser for the Git index file to accurately manage the staging area, track file metadata, and detect worktree modifications.
- Resolved object identity discrepancies by writing custom lexicographical sorting algorithms for tree objects, ensuring perfect compatibility with the official Git CLI across 15 core commands.

Taskly - Kanban App with AI Assistant - *React, Supabase, Vercel* | GitHub | Live Demo — Apr. 2026

- Diagnosed and solved a 60fps scroll regression caused by expensive CSS effects, rebuilding the HTTP layer with Node.js native HTTPS and passive event listeners to improve touch navigation performance.
- Eliminated cross-tab data inconsistencies by designing a centralized mutation pipeline with an undo/redo history and a 500ms-debounced Supabase sync, maintaining a single source of truth across guest and authenticated modes.
- Integrated Groq's DeepSeek-R1 70B to parse natural-language commands, utilizing a regex-fallback JSON extraction method to handle unpredictable LLM outputs and resolve fuzzy task/column queries.

Experience & Leadership

SCHOOL MUN — In-person
Head of IT — *Apr. 2024 – Feb. 2025*

- Directed IT logistics and media operations for a 2-day student-run Model UN conference, managing AV infrastructure, livestreaming, and registration systems across multiple committee rooms.

Clevered AI Fellowship — Remote
Student Fellow — *Jan. 2024 – Feb. 2024*

- Selected for a mentored AI program led by Oxford-affiliated researcher Ken Kahn; built a voice-activated accessibility prototype in Python using NLP-style intent matching to handle varied user commands.

Achievements

Bronze Medalist: TIMO International Math Olympiad
Finalist: Northeastern University London Essay Competition (Tech & Society)